

Not everything is concrete

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Purpose of the talk

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To do this we will need categories that are not concrete.

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- ▶ *Axioms*

$(h \circ g) \circ f = h \circ (g \circ f)$ and $f \circ 1_A = f = 1_B \circ f$.

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In all these examples the morphisms are functions, possibly with a property. All these categories are 'concrete' in a sense we will define precisely below.

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In these two examples the morphisms are literally not functions.

However, **Rel** is concrete since a relation $R \subseteq A \times B$ can be regarded as a union-preserving function $PA \longrightarrow PB$ between the powersets.

That **Hot** is not concrete is a theorem of Peter Freyd.

An example from knot theory

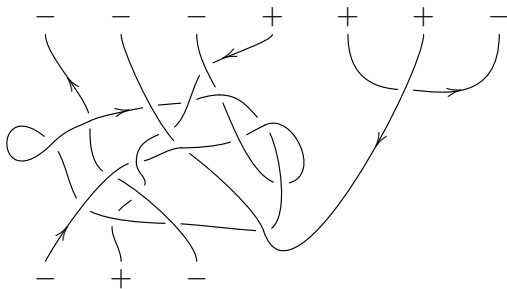
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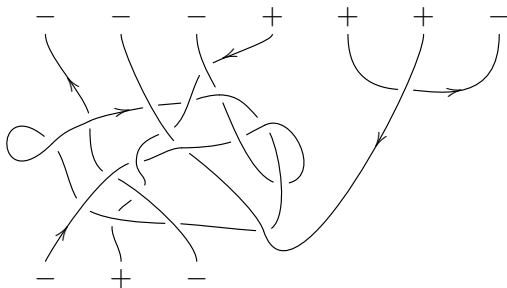
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- ▶ Composition is vertical gluing of tangles

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There is a category **Cat** whose objects are (small) categories and whose morphisms are functors.

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A category $\mathcal{A}$ is called *concrete* when it is equipped with a faithful functor $\mathcal{A} \longrightarrow \mathbf{Set}$.

The opposite of a category

For each category $\mathcal{A}$ there is a *dual* or *opposite* category denoted by $\mathcal{A}^{\text{op}}$ defined by:

- ▶ the objects are the same as for $\mathcal{A}$
- ▶ the morphisms are the same as for $\mathcal{A}$ with domain and codomain reversed; so $f : A \longrightarrow B$ in $\mathcal{A}$ becomes $f : B \longrightarrow A$ in $\mathcal{A}^{\text{op}}$
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Here is a question answered for some $\mathcal{A}$ by classical theorems:

¿If $\mathcal{A}$ is concrete, is $\mathcal{A}^{\text{op}}$?

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- ▶ **Rel**^{op} is equivalent to **Rel** (and so is concrete)
- ▶ **Tang**^{op} is equivalent to **Tang** (but is not concrete)

Monoidal categories

Many categories $\mathcal{A}$ have a natural choice of functor

$$- \otimes - : \mathcal{A} \times \mathcal{A} \longrightarrow \mathcal{A} ,$$

which is associative up to canonical isomorphism

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With a couple of axioms, this defines *monoidal category*.

Examples of monoidal categories

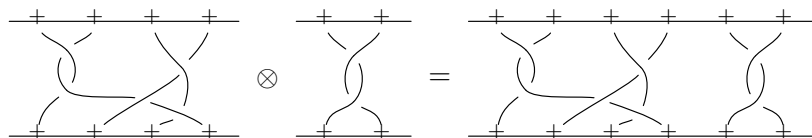
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- ▶ **Vect** has ordinary tensor product $V \otimes W$ of vector spaces with $I = \mathbb{C}$, the complex numbers
- ▶ **Tang** has horizontal stacking of tangles with I the empty word in the symbols $+$ and $-$

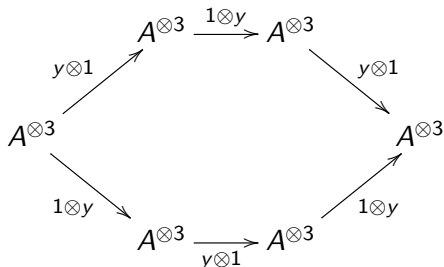


Yang-Baxter (YB) operators

A *Yang-Baxter operator* on an object A of a monoidal category $\mathcal{A}$ is an invertible morphism

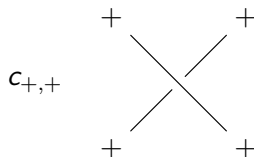
$$y : A \otimes A \longrightarrow A \otimes A$$

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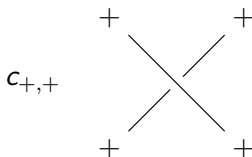
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- ▶ A YB-operator R_q on $V \in \mathbf{Vect}$ (for $q \in \mathbb{C}$ and $e_1, \dots, e_n$ a basis of V) is defined by:

$$R_q(e_i \otimes e_j) = \begin{cases} e_j \otimes e_i & \text{for } i > j \\ e_j \otimes e_i + (q - q^{-1})e_i \otimes e_j & \text{for } i < j \\ qe_i \otimes e_j & \text{for } i = j \end{cases}$$

Link invariants and quantum groups

- ▶ A tensor-preserving functor $F_q : \mathbf{Tang} \longrightarrow \mathbf{Vect}$ is defined on objects by using the indicative formula

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- ▶ The endomorphisms of $\mathbb{C}$ in **Vect** are complex numbers.
- ▶ **Link invariant**: Each link L is assigned a complex number $F_q(L)$.
- ▶ **Quantum group**: A co-endomorphism construction from Tannaka duality theory yields a Hopf algebra $\text{End}^\vee(F_q)$.